

Paper 2.75 - Correction Surface as Next-State Precondition

CQE-paper-02.75

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-02.75.md

Paper 2.75 - Correction Surface as Next-State Precondition

Purpose

Paper 2.75 explains how Paper 2 becomes a controlled precondition for Papers 3 and 4.

Exported State

Paper 2 exports:

```
the LCR carrier from Paper 1
the Rule30 / Rule90 comparison
the correction term C and not R
the firing states (0,1,0) and (1,1,0)
the D4 coordinates (2,0) and (3,1)
the rule that residue is obligation, not proof
```

Use in Paper 3

Paper 3 may use the two firing states as registered chart entries only if it preserves:

```
the original LCR state
the correction value
the D4 axis/sheet coordinate
the receipt that proves the row
```

Paper 3 may prove stronger registration results, but those results must be proved there. Paper 2 exports coordinates; it does not prove the whole triality surface.

Use in Paper 4

Paper 4 may treat the nonzero correction rows as boundary repair inputs. It may not erase the fact that the rows began as failed-route residue.

The allowed transition is:

```
correction residue -> typed repair constraint
```

The disallowed transition is:

```
correction residue -> closed proof without repair
```

Use in Later Papers

Later papers may lift the correction surface into:

- triality/Jordan registration,
- boundary repair,
- rolling path payloads,
- causal graph receipts,
- sampling and interpolation surfaces,
- Hamiltonian or mass-residue carriers.

Every lift must name which additional theorem closes the new structure.

Precondition Rule

Before a later paper uses Paper 2, it should be able to answer:

Which LCR state is active?
What is the correction value?
What route failed or required correction?
Is the residue closed, open, or transported?
Which receipt proves the imported row?

Conclusion

Paper 2.75 turns the correction surface into a next-state input. It keeps the residue visible as the suite moves from correction to registration and then to boundary repair.